

CHENMING GE

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EDUCATION

University of Michigan, Ann Arbor
B.S.E. in Computer Science

Ann Arbor, USA
Aug 2025 - Present

- **GPA:** 3.88/4.00
- **Selected Coursework:** Machine Learning, Operating Systems, GPU Parallel Programming, Database Management Systems.

Shanghai Jiao Tong University
B.Eng. in Mechanical Engineering

Shanghai, China
Aug 2023 - Present

SELECTED RESEARCH PROJECTS

WebEvolve: Experience Robustness for Web Agents Under Version Drift
Advisor: Prof. Honglak Lee

Ann Arbor, MI
Jan 2026 – Present

- Investigating how **MLLM-based web agent memory** degrades under **website version drift**, proposing **Experience Transfer Gap** metrics to quantify experience decay across a **7-category drift taxonomy**.
- Implemented **ExpeL insight extraction** and **Agent Workflow Memory** induction pipelines on collected agent trajectories, comparing robustness of different **experience abstraction levels** (trajectory, insight, workflow) to controlled drift.
- Exploring **LLM-synthesized (vibe-coded) web environments** with known, injected drift as a controllable testbed; designed a **structural and functional verification pipeline** to validate that synthesized sites preserve realistic interaction semantics.
- Built the underlying benchmark infrastructure: **Docker-containerized multi-version environments**, a **knowledge-graph-based cross-version diff engine**, and an automated **task maintenance pipeline** within BrowserGym and AgentLab.

Efficient Inference for Embodied Foundation Models
Advisor: Dr. Jiachen Liu

Ann Arbor, MI
Nov 2025 – Present

- Conducted **lesion-based mechanistic analysis** on Cosmos Policy, a 2B-parameter video-generation DiT repurposed for robot control, across LIBERO and RoboCasa benchmarks.
- Discovered that **action-critical and video-critical computation are mechanistically decoupled**: action fidelity concentrates in early transformer blocks while future-frame prediction resides in later blocks, an invariant that holds across benchmarks.
- Leveraging these interpretability findings to design **block-selective caching and token pruning** strategies that skip redundant visual computation while preserving action quality.

Probabilistic Motion Planning for Redundant Robots
EECS 465 (Intro to Algorithmic Robotics) course project

Ann Arbor, MI
Nov 2025 – Dec 2025

- Evaluated RRT-Connect and PRM on a 7-DOF Franka Panda; optimized sampling strategies in high-dimensional state spaces, achieving a 75% reduction in planning latency.

Gravitational Effect on Swarming Behavior of Microorganisms
Advisor: Prof. Zijie Qu

Shanghai, China
Sep 2024 - Aug 2025

- Trained a **U-Net** model for colony boundary segmentation; diagnosed systematic failure modes, then adopted a **SAM-based** pipeline that achieved reliable automated detection, replacing manual annotation.

SKILLS

Frameworks	PyTorch, Hugging Face, CUDA, Docker, Linux
Languages	Python, C/C++, Java, SQL, LaTeX, Typst